



Project Requirement Document (PRD)

Overview

PROJECT TITLE: SAHAYAK

I. Target Group (Who ?)

- **Elderly People** - They often find it difficult to handle technology and get comfortable with it.
- **People with Disabilities** - Our app can be used by anyone that suffers from conditions like visual impairment, hearing loss, or mobility issues.
- **Technologically Inexperienced Users** - People who face difficulty in using smartphones and new technology.
- **Low Literacy Users** - People who find it difficult to read and write can also use SAHAYAK.

DEMOGRAPHICS	PSYCHOGRAPHICS
● Age: 13 and above ● Gender: Male, Female, Non Binary and Others ● Region: India	● Wishes to connect with their family. ● Wants to feel independent and empowered. ● Wants easy and accessible technology.

User Personas



"I just wish my phone was simpler, so I could call my daughter without asking for help every time."

Name: Ramprasad Sharma
Age: 78 years old
Occupation: Retired School Principal
Location: Varanasi, Uttar Pradesh

User Information:
In recent years, Ramprasad has started experiencing mild cognitive decline, leading to forgetfulness and difficulty navigating his smartphone.

Frustrations:

- Confused by new smartphone features.
- Difficulty remembering app locations.
- Discomfort with not being as sharp as before.

Personal Characteristics:

- Respects routine and structure.
- Eager to stay mentally active and up to date.
- Self-reliant but now feels uncertain.

Needs:

- Memory aids for smartphone navigation.
- Voice prompts to guide app usage.
- Clear reminders for important tasks.

Goals:

- Stay mentally sharp and engaged.
- Maintain independence in managing daily tasks.
- Regain confidence in using technology

Technology Confidence:

- Technology Familiarity: 50%
- App Navigation Confidence: 25%
- Willingness to Learn: 90%



Name: Bhagirathi Devi

Age: 82 years old

Occupation: Homemaker

Location: Madhubani, Bihar

User Description:

Bhagirathi Devi is a respected elder in her village and is known for her wisdom and herbal remedies.

However, her deteriorating eyesight due to ageing makes it difficult for her to dial numbers, read messages or even recognise contacts on her phone. She relies on her son to help her with phone related tasks.

Personal Characteristics:

- Highly respected in her village.
- Prefers traditional methods.
- Relies on family support.

Feelings:

- Feels overwhelmed by technology.
- Longs for more family interaction.
- Comforted by familiar surroundings.

Goals:

- Stay connected with family.
- Be independent in communication.
- Learn to use technology.

Frustrations:

- Difficulty in using smartphones.
- Frustrated by complex app interfaces.
- Feels disconnected from distant family.

Needs:

- Simple, easy to use technology.
- Larger, clear text and buttons.
- Frequent connection with family.

Technology Confidence:

- Low Confidence- She is hesitant and feels overwhelmed by modern technology.



User Information:

Aditi is a high school student who loves spending time with her grandparents. However, she finds it difficult to stay in touch with them because they struggle to use their smartphones. She wishes they could reply to her texts and video call her more easily without needing help from others.

Personal Characteristics:

- Empathetic and supportive.
- Tech-savvy but patient with others.
- Friendly and approachable.
- Organised and goal oriented.
- Curious and eager to learn.

Name: Aditi Mehra

Age: 15 years old

Occupation: High School Student

Location: Mumbai, Maharashtra

Goals:

- Stay connected with her grandparents easily.
- Help her grandparents use technology easily.
- Empower her grandparents to use smartphones confidently.

Pain Points:

- Difficulty in teaching her grandparents to use tech.
- Frustrated by her grandparents' fear of technology.
- Limited tools for facilitating easy communication with them.
- Worry about her grandparents' safety.

Needs:

- Simple tech solutions for communication with her grandparents.
- Voice-assistance features to guide her grandparents.
- Tech that provides real-time help for her grandparents.



Demographic Information:

Name: Sunita Rao
Age: 45 years old
Occupation: Literature Prof.
Location:
Bangalore, Kamataka

User Information:

Sunita, a literature professor, has been living with a mild motor impairment that affects her dexterity. While she is tech-savvy, prolonged smartphone usage strains her fingers, making typing and navigation difficult. She wishes for a more accessible solution that reduces her need for precise touch gestures.

Personal Characteristics:

- Proud of her independence despite mobility challenges.
- Community-oriented and social.

Hobbies and Interests:

- Listening to music, especially classical.
- Watching TV shows and movies
- Spending time with her children.

Goals:

- Maintain independence in her daily tasks.
- Stay connected with family and friends.
- Easily use technology for communication.

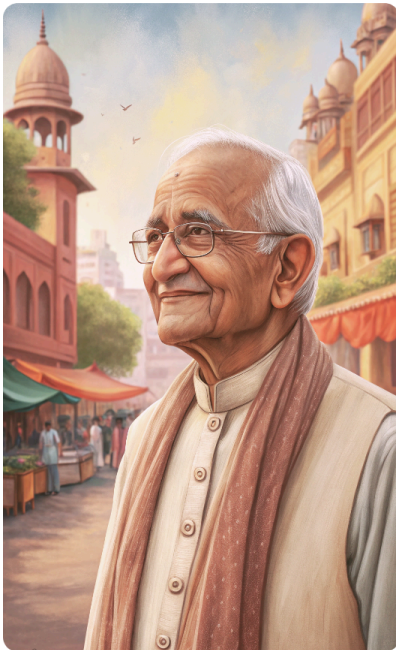
Challenges:

- Difficulty with certain motor tasks on smartphones
- Occasional frustration with small screen size and buttons.

Needs:

- Voice assistance for hands-free tasks.
- Simplified communication options, like video calls.
- Accessible tech with larger buttons and simple UI.

"It's so frustrating when I can't send a message properly; I need something simpler."



Demographic Information:

Name: Gopal Das
Age: 75 years old
Location: Jaipur, Rajasthan
Occupation:
Retired Government Clerk

User Information:

A retired government clerk, Gopal Das never had to use a smartphone extensively. He knows how to answer call but struggles with texting, saving contacts, or using apps.

Personal Characteristics:

- Prefers face to face interactions.
- Relies on his grandson for technological help.
- Traditional and set in his ways.

Goals:

- Connect with old friends independently.
- Easily make calls and send messages.
- Stay in touch with family without help.

Pain Points:

- Struggles with reading and writing.
- Feels embarrassed asking for tech help.
- Finds it difficult to relate to his much younger grandson.

Needs:

- Assistance with basic phone functions.
- Simple UI with large buttons.
- Voice based call and messaging features in native language.

II. User Story: A Customer Journey

Ommati Devi's Journey: From Struggle to Independence



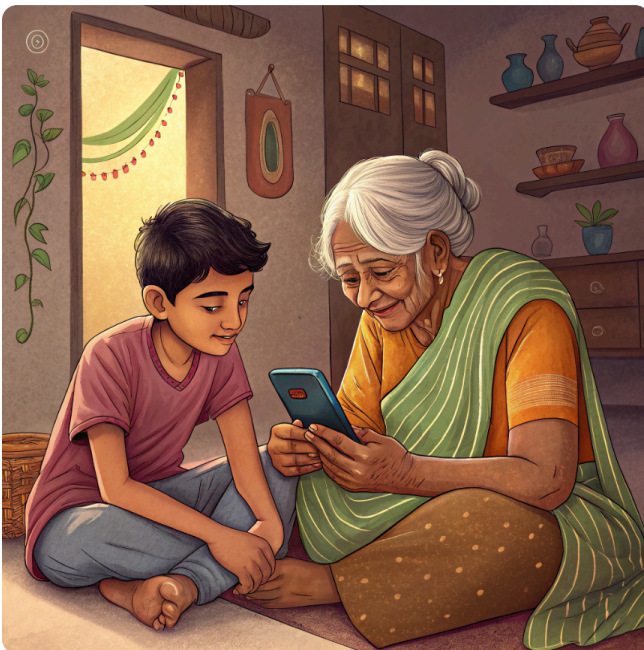
Ommati Devi is a **72-year-old woman** living in a small town in India. She has spent most of her life managing her household and caring for her family. Though she never had the need to use a phone extensively before, her daughter, who now lives abroad, gifted her a smartphone to stay connected. Ramkali was excited at first, as her daughter taught her how to make video calls and send messages. However, **due to her age and unfamiliarity with technology**, she quickly forgot how to use it.

One day, Ommati feels the urge to speak to her daughter. She picks up the phone but **cannot recall** how to find her contact. She taps a few buttons but ends up opening the wrong apps. **Frustrated**, she sighs and puts the phone aside, feeling helpless. She considers asking her grandson for help, but he is often busy, and **she doesn't want to disturb him**.





As days pass, she begins to **feel disconnected**. She misses her daughter deeply but **doesn't want to depend on others** for something as simple as making a call. Every time she needs to use her phone, **she hesitates**, fearing she might press the wrong button or delete something important. This growing frustration makes her **feel increasingly isolated**.



One afternoon, her grandson notices her struggle and **installs SAHAYAK** on her phone. He explains that she can now use simple **voice commands to make calls, send messages, and even listen to texts** without navigating complicated menus. Curious but skeptical, Ommati gives it a try. "**Call my daughter,**" she says hesitantly. To her surprise, the phone immediately dials her daughter's number. A wave of relief washes over her as she hears her daughter's voice on the other end.

Over time, Ommati grows more confident with her phone. She no longer waits for someone to help her; instead, she easily **calls her daughter whenever she wants**. She listens to messages **without struggling** to read small text and responds effortlessly. SAHAYAK has given her the **independence** she longed for, turning what was once a source of frustration into a tool of empowerment. Now, Ommati enjoys her calls with her daughter, **feeling connected and in control again**.



III. Why this Project (Why ?)

Smartphones offer assistive features such as simplified user interfaces and voice agents like Siri or Google Assistant, which can perform tasks through voice commands. However, many elderly individuals, people with disabilities, and those with low literacy skills, especially in the Indian context struggle to navigate modern smartphones.

They often rely on others for daily phone usage or even fear using these devices altogether. Common challenges include:

1. Making or receiving calls
2. Reading and replying to messages
3. Capturing images
4. Watching videos on platforms like YouTube, and
5. Combating feelings of isolation and helplessness.

Handling technology can be difficult due to various factors such as **cognitive or motor impairments, low literacy, upbringing, and environment**. These challenges are often unaddressed and create significant barriers to digital inclusion. **SAHAYAK** aims to bridge this gap by providing a simple and accessible way for users to stay connected with their families. The project focuses on developing an inclusive smartphone application that simplifies essential tasks using a multimodal interaction system incorporating voice, gestures, and a streamlined UI, making technology more accessible and empowering for its users.

IV. Our Plan (What ?)

Our Plan for SAHAYAK is to provide the following core functionalities:

Calling	● Large, easy-to-press buttons. ● Voice-based calling (e.g., "Call my daughter"). ● Emergency contacts with one-touch access.
Messaging	● Voice-assisted message composition and reading. ● Option to read messages aloud.
Contact Management	● Voice-based contact saving (e.g., "Save the last called number as Sandeep"). ● Simple, categorized contact lists. ● Saving the numbers with images if obtained from WhatsApp or Gmail scarping.
Call Answering	● Answering calls via proximity (e.g., bringing the phone close to the ear). ● Voice-based call acceptance (e.g., "Pick up the call").
Fall Detection	● Integrating motion sensors to detect falls and trigger emergency alerts.
AI Assistance	● A virtual assistant to answer questions and provide general help. ● Activated via a single button press or wake word.
Multilingual Support	● Speech-to-text and text-to-speech capabilities in multiple languages. ● Real-time translation to assist in communication.

V. Existing Relationships and Needed Changes

Master-Slave (Needs to Change): Many elderly or disabled users rely on family members or caregivers to handle their phones, making them dependent. This project aims to remove that dependency and empower users to take control of their devices. Instead of technology dictating their actions, it should seamlessly work with them in an easy and natural way.

Partner-Partner (Should Stay): The app is designed to be a helpful teammate, not a source of frustration. By incorporating voice commands, simple buttons, and smart AI, **SAHAYAK** becomes a supportive partner that assists users rather than confusing or overwhelming them.

Trusted-Advisor (Should Stay & Improve): The **AI assistant in SAHAYAK** acts as a friendly guide, answering questions and assisting with tasks like saving contacts or sending messages. Building a trust-based relationship is essential, which is why the AI must understand different languages, accents, and user needs to provide a seamless experience.

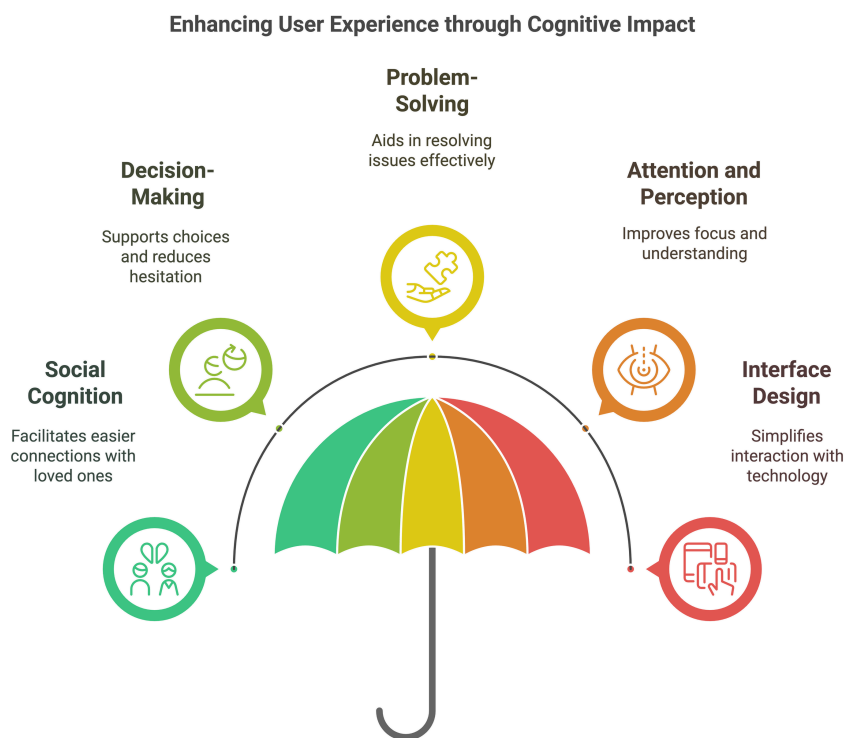
What’s Changing and What’s Staying the Same?

Changing: Moving away from complicated smartphone interactions and reducing user dependency on others.

Keeping & Improving: Ensuring that SAHAYAK acts as a helpful teammate and a trustworthy guide so users feel confident using their phones without frustration.

VI. Cognitive Impact on Users

The app enhances social cognition by making it easier for users to connect with their loved ones. It supports decision-making and problem-solving, reducing hesitation and anxiety when using technology. By leveraging multi-modality, it improves attention and perception, ensuring a more accessible experience. Additionally, the intuitive and simple interface minimizes the need for learning and memory retention, making interactions effortless.



VII. Our Value Proposition

Empowerment & Independence	Use technology without reliance.
Simple & Familiar Design	No complexity, just ease.
Multi-Modal Interaction	Voice, gestures, and intuitive UI.
Safety & Security	Emergency contacts and fall detection.
Bridging the Digital Divide	Making technology truly accessible.

VIII. How ?

Innovative Features for Inclusive and Safe SAHAYAK App Development



NLP Integration

Developing speech recognition and intent interpretation for voice commands.



Multimodal Interaction

Supporting touch, voice, and gesture methods for flexible user engagement.



User-Friendly Interface

Designing a simple, high-contrast interface for easy navigation.



Multilingual Support

Implementing automatic speech translation and text-to-speech in multiple languages.



Sensor-Based Safety

Utilizing smartphone sensors to detect falls and notify emergencies.

IX. Impact SAHAYAK has on User Satisfaction, Emotion and Trust

1. Satisfaction– Feeling Independent

- Users can make calls, send messages, and navigate their phones **without needing help**.
- Simple design and voice commands reduce frustration and make daily tasks easier.
- Emergency contacts and fall detection provides peace of mind and safety.

Impact: Users feel more confident and capable of using their phones on their own.

2. Emotions– Feeling Connected and Comfortable

- Many elderly people or individuals with some disability **feel left out** because they struggle with their smartphones.
- **SAHAYAK** helps them **navigate** through their smartphone with much ease and **stay connected** with their family through easy calling and messaging.
- The frustration of using complicated interface is replaced with a sense of **ease, comfort and empowerment**.

Impact: Users feel less lonely and more comfortable using technology.

3. Trust– Feeling Safe and Supported

- The app is designed to be **helpful, not confusing**.
- The voice assistant provides **clear and simple guidance**.
- Features like large buttons and multilingual support show that the app **understands their needs**.

Impact: Users trust the app because it feels reliable and easy to use.

